

KUNAL M SHENDE

Embedded Systems Engineer | IoT Security Researcher

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PROFESSIONAL SUMMARY

Computer Science undergraduate with 2+ years of experience in embedded systems, IoT hardware security, and web development. Proficient in vulnerability assessment, penetration testing, and firmware engineering on ESP32-S3 and ATmega328P platforms. National competition award winner and Indian Design Patent holder.

TECHNICAL SKILLS

Programming Languages: C, Embedded C, C++, Python, TypeScript, Bash

IoT and Embedded Hardware: ESP32, ESP32-S3, Arduino, Raspberry Pi, FreeRTOS, SPI, I2C, UART

Web Development: HTML5, CSS3, Bootstrap, Python Flask, WebSockets, Firebase

Databases: MySQL, Supabase, Firebase Firestore

Cybersecurity Tools: Burp Suite, Nmap, Wireshark, Metasploit, OWASP Top 10

DevOps and Cloud: Git, GitHub, Docker, Vercel, Google Cloud Platform, VS Code, Postman

WORK EXPERIENCE

Independent Security Researcher | Freelance | January 2024 - Present

- Architected 3 secure IoT prototypes on ESP32-S3 and ATmega328P, reducing firmware attack surfaces by 40% with memory-safe coding practices.
- Discovered 15+ critical vulnerabilities in 10 controlled environments using OWASP and NIST assessment frameworks.
- Mitigated network anomalies by 25% via packet-level traffic analysis and proactive system hardening across 5 IoT deployments.

Web Exploit Hunting Intern | EduSkills and AICTE | June 2026 - August 2026

- Detected 12 security flaws including SQL injection and XSS within 8 web applications during structured penetration testing exercises.
- Compiled detailed vulnerability reports with severity ratings, accelerating remediation timelines by 30% for the development team.

Cybersecurity Intern | Palo Alto Networks and EduSkills | January 2026 - March 2026

- Analyzed 500+ packet captures and firewall rule sets, strengthening endpoint protection for 3 cloud environments.
- Earned 4 industry certifications from Palo Alto Networks in Security Operations, Network Security, Cybersecurity, and Cloud Security.

Java Full Stack Developer Virtual Intern | EduSkills and AICTE | October 2025 - December 2025

- Completed a 12-week program covering full-stack application development, relational database integration, and backend service design.

Cloud Virtual Intern | AWS Academy | April 2025 - June 2025

- Studied 10+ core AWS services including EC2, S3, VPC, and IAM over an 8-week hands-on lab curriculum.

Android Developer Virtual Intern | Google for Developers | January 2025 - March 2025

- Explored 8 Android modules on Material Design, activity lifecycle management, and local SQLite data persistence.

PROJECTS

ZeroTrace - Cybersecurity Audit Device | ESP32-S3, FreeRTOS, C++, USB-HID | [\[GitHub Repo\]](#) | 2026

- Engineered an embedded audit companion integrating 4 modules: RF scanning, Wi-Fi auditing, RSSI foxhunting, and USB-HID injection.
- Optimized display rendering to 60 FPS by pinning FreeRTOS tasks on dual CPU cores, cutting frame latency by 35%.
- Secured 2nd Prize at TECHTANTRA 2K26 national competition among 50+ teams.

Sentinel - HID Attack Detection Framework | Python, Windows API, WMI | 2026

- Developed a real-time USB threat detection system that neutralizes rogue HID devices within 10 milliseconds.
- Intercepted 100% of simulated BadUSB attack vectors via automated forensic snapshotting in controlled scenarios.

MicroFlasher - Web Serial Flashing Platform | *TypeScript, Web Serial API* | 2026

- Deployed a zero-install browser-based firmware flasher, reducing developer onboarding time by 70%.

Autonomous Line-Follower Robot | *ATmega328P, L298N, IR Sensors* | [\[GitHub Repo\]](#) | 2025

- Constructed a differential-drive robot with PWM calibration at 7.8 kHz, eliminating motor stalling and reducing noise by 90%.
- Calibrated a proportional control algorithm achieving 95% path accuracy on sharp 90-degree corners.

ACHIEVEMENTS

- Won 2nd Prize at TECHTANTRA 2K26 National Level Technical Competition among 50+ teams for the ZeroTrace project.
- Filed Indian Design Patent (Application No. 499678-001, CBR No. 209757) in April 2026 for ZeroTrace hardware.

EDUCATION

Bachelor of Technology in Computer Science and Engineering | *S.B. Jain Institute of Technology, Nagpur* | August 2023 - June 2027
Specialization: IoT Architecture, Cybersecurity, Embedded Systems

Higher Secondary Certificate (HSC), Science | *Municipal Junior College, Kalmeshwar* | June 2021 - March 2023

Secondary School Certificate (SSC), Distinction | *Municipal Junior College, Kalmeshwar* | June 2019 - March 2021

CERTIFICATIONS

- Cyber Security Workshop** - *Crow Security and IIT Bombay*, 2025-2026
- Ethical Hacking and Penetration Testing** - *NASSCOM FutureSkills Prime*, 2025-2026
- Security Operations Fundamentals** - *Palo Alto Networks*, March 2026
- Network Security Fundamentals** - *Palo Alto Networks*, March 2026
- Cybersecurity Fundamentals** - *Palo Alto Networks*, March 2026
- Cloud Security Fundamentals** - *Palo Alto Networks*, March 2026
- Python Basic and Problem Solving** - *HackerRank*, 2025
- AWS Cloud Computing Fundamentals** - *AWS Academy*, 2025
- German Language Proficiency Level 1** - *FLOA*, December 2025

LANGUAGES

English (Professional) | Hindi (Native) | Marathi (Native) | German (Beginner)